

Proposal for Women in Engineering & Entrepreneurship 12-Hour Hackathon at Hack MIT-WPU

Introduction

The **Women in Engineering & Entrepreneurship 12-Hour Hackathon** at Hack MIT-WPU is more than just a competition, it's a movement to break barriers and champion the next generation of female innovators. This high-energy, fast-paced event will provide participants with hands-on experience in building groundbreaking engineering solutions and entrepreneurial ventures, fostering creativity, problem-solving, and collaboration. It's designed to be a launchpad for women to redefine industries, disrupt norms, and create the future of tech and business, all in just 12 hours!

What makes this track stand out?

⌘ Intensive 12-Hour challenge designed to simulate real-world problem-solving under time constraints.

- ☐ Get guidance from top women leaders in tech and business, offering insights, feedback, and career advice.
- ☐ Address issues specifically impacting women in STEM, finance, healthcare, and safety.
- ☐ Special awards, investment opportunities, and post-hackathon incubation support for standout projects.
- ☐ Engage with like-minded peers, industry experts, and potential investors in a supportive ecosystem.

Hackathon Structure

This hackathon is designed to empower women in technology and business by encouraging them to develop real-world solutions within a condensed 12-hour timeframe. Participants will combine technical expertise with business acumen to create innovative, scalable ventures. Collaboration, leadership, and skill development take center stage, ensuring every participant walks away with a valuable, hands-on experience.

Hackathon Components

1. Problem Statements:

Participants will tackle thought-provoking challenges that push them to develop engineering solutions or startup ideas with real-world applications, all within 12 hours.

2. Solution Development and Integration:

Teams will engage in the rapid design, development, and integration of solutions using cutting-edge engineering principles and entrepreneurial strategies. The goal is to create sustainable, scalable, and innovative products or services in a short yet impactful timeframe.

3. Judging Criteria:

Solutions will be evaluated based on:

- **Technical Feasibility** – Engineering depth and execution.
- **Creativity & Innovation** – Originality and uniqueness of the idea.
- **Business Viability** – Market readiness and scalability.
- **Social Impact** – Effectiveness in addressing real-world challenges.

Sample Problem Statements:

Finance & Economic Empowerment

1. **Equal Pay Transparency Tool** – Build a blockchain-powered tool that promotes salary transparency across industries and helps women negotiate fair pay.
2. **Financial Literacy for Women** – Create an AI-powered app that helps women manage finances, understand investments, and close the gender wealth gap.
3. **Women-Owned Business Marketplace** – Develop an online platform that connects women entrepreneurs with potential investors, mentors, and customers to scale their businesses.

Safety & Security

1. **AI-Powered Harassment Detection** – Design an AI tool that can detect and report workplace harassment through email, chat, or video call analysis.
2. **Emergency Response Network for Women** – Develop a crowdsourced real-time safety app where women can request immediate assistance from nearby verified responders.
3. **Cybersecurity for Women** – Create a tool that educates women on protecting themselves against online harassment, doxxing, and financial fraud.

STEM Education & Career Advancement

1. **AI-Based STEM Mentorship** – Build an AI-powered mentorship platform that matches women in STEM with mentors, tracks their career progress, and offers personalized learning resources.
2. **Bias-Free Hiring in Tech** – Develop an AI-powered resume screening tool that removes gender biases from hiring decisions.
3. **EdTech for STEM Upskilling** – Create an interactive learning platform with AR/VR to teach STEM concepts to young girls and bridge the gender gap in STEM fields.

Healthcare & Well-being

1. **Accessible Women's Health Diagnostics** – Develop an affordable, non-invasive diagnostic tool for early detection of diseases like PCOS, endometriosis, or cervical cancer.
2. **AI for Mental Health** – Create an AI-powered chatbot or platform that provides mental health support tailored to women's unique challenges, including workplace stress, postpartum depression, and harassment trauma.
3. **Wearable Tech for Women's Safety** – Design a smart wearable device that can detect distress and alert emergency contacts with real-time location tracking and audio recording.
4. **Remote Prenatal & Postnatal Care** – Build a telehealth platform that supports women in remote areas with prenatal and postnatal care, offering virtual consultations and AI-driven health monitoring.

Smart Assistive Technologies

1. **Inclusive Design Challenge** – Develop an interactive AI-powered accessibility tool that assists differently-abled individuals in STEM fields. It could include voice-controlled coding assistants, AR-based lab simulations, or real-time transcription tools.
2. **Wearable Health Monitor for Women** – Design a smart wearable device that tracks women's health indicators (such as stress levels, sleep patterns, and hormonal changes) and provides actionable insights.

Future of Work & Leadership

1. **AI-Powered Career Coach** – Build a virtual mentor that provides personalized career growth strategies, including resume feedback, mock interviews, and networking guidance for women in STEM.
2. **Bias-Free Performance Evaluation System** – Design an interactive tool that uses AI to evaluate employee performance based on skills and impact rather than gender-biased factors.

Sustainability & Social Impact

1. **Eco-Innovation for Women Entrepreneurs** – Create a platform that connects women-led businesses with sustainable resources and eco-friendly manufacturing options.
2. **Gamified Sustainability Challenges** – Develop an interactive app where teams can compete to complete eco-friendly tasks (e.g., carbon footprint reduction, zero-waste

solutions, energy-efficient business models).

Cybersecurity & Online Safety

1. **AI-Based Scam Detector** – Build a tool that scans financial transactions, job offers, or social media messages to detect fraud and alert users in real time.
2. **Interactive Cyber Hygiene Training** – Develop a game-based cybersecurity awareness program that teaches essential online safety practices through interactive storytelling.

Objectives

- **Encourage Women in STEM & Business** – Create an inclusive space for women to explore technology and entrepreneurship.
- **Develop Real-World Solutions Quickly** – Foster innovation by solving pressing challenges within a 12-hour timeframe.
- **Hands-On Experience** – Enable participants to work with AI, IoT, and business frameworks in a high-pressure environment.
- **Industry Exposure** – Connect participants with experts, mentors, and investors.

Outcome

- **Skill Enhancement** – Participants will gain technical and business skills applicable in real-world scenarios.
- **Portfolio Projects** – Teams will build projects that can be showcased for career or startup opportunities.
- **Networking & Mentorship** – Access to a network of industry leaders, investors, and experts.

- **Empowerment & Recognition** – Women-led teams will receive support to further develop their ideas post-hackathon.